Radar Altimeter System (Mod 6/2049) – General Data

1. General

The radar altimeter system RAD ALT 1 is part of the basic aircraft avionics suite, (Mod 6/2049). Provisions for a second radar altimeter system RAD ALT 2 have been made and is installed as customer option order 6218.

2. Description

The KRA 405B Radar Altimeter (RAD ALT) transceiver is installed in the empennage aft of the rear baggage compartment. A second radar altimeter can be fitted if desired. The transceiver sends a signal to the transmit antenna and gets the return signal from the receive antenna. The maximum operating height Above Ground Level (AGL) used by the system is 2,500 feet. The radar altimeter system measures the aircraft height AGL electronically and sends the height AGL data to the Modular Avionics Unit (MAU) for display in the ADI window of the left and right PFDs. The digital readout for radio altitude is displayed in white text in a black box in the lower center part of the attitude display on the PFD. The radar altitude display is removed at altitudes greater than 2,500 feet. When altitude is less than 550 feet, the lower portion of the PFD altitude tape will show a yellow cross hatched box to indicate the ground proximity. If the radar altitude data becomes invalid, the digital readout will be replaced with RAD in white in an amber box. The radar altimeter data (in addition to indicated airspeed data from the ADAHRS) is used by the aircraft air-ground monitor logic system to determine on ground or in-flight status.

The radar altimeter is an integral component of the Class A TAWS (Terrain Awareness and Warning System) and must be installed in aircraft that are ordered with TAWS.

RAD ALT 1 receive antenna is installed on the bottom of the rear fuselage at FS416 and the transmit is located at FS431. RAD ALT 2 wire integration, circuit breaker, and mounting tray location are provisioned in the aircraft.

28 Volt DC power is protected by a 3 Amp RAD ALT 1 (Left Bus) and a provisioned RAD ALT 2 (Right Bus) circuit breaker, located on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

The KRA 405B Radar Altimeter System interfaces to the following aircraft system:

A. Modular Avionics Unit (MAU) (refer to 31-43-00, Modular Avionics Unit (MAU) (Mod 6/2049) – General Data).

B. MK VI Enhanced Ground Proximity Warning System (EGPWS) (refer to 34-44-00, Enhance Ground Proximity Warning System (TAWS) (Mod 6/2049) – General Data), if installed.

C. Terrain Collision Avoidance System (TCAS) (refer to 34-43-00, Traffic Alert and Collision Avoidance System I (TCAS I) – General Data and 34-43-01, Traffic Alert and Collision Avoidance System II (TCAS II) (Mod S.O.O. 6219) – General data), if installed.

3. Adjustment/Test

A. Radar Altimeter Transceiver (KR 405B)

1) General

These procedures are for the adjustment and test, and the height calibration procedures for RAD ALT 1 Transceiver and for RAD ALT 2 Transceiver, if installed.

2) Testing of RAD ALT Transceiver (KR 405B)

a) Ensure that RAD ALT 1 (J5) or RAD ALT 2 (J6) circuit breaker on the avionics circuit breaker panel is closed.

b) Inject a DC variable power supply (0 to 25 Volt DC) signal into the transceiver test input. Ensure correct polarity before turning the power supply on.

c) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

d) Wait several minutes for the aircraft electrical and avionics systems to initialize.

e) On the Flight Control Panel, use the MINIMUMS knob to set Radar Altimeter (RA) Decision Height (DH) to 50 ft.

f) Slowly increase the DC variable power supply voltage until 150 feet is displayed.

g) Connect the Air Data Test set to the aircraft pitot ports and increase airspeed to > 55 Knots.

h) Ensure aircraft changes from “Ground” to “Air” mode.

i) Wait 60 seconds.

j) On the Flight Control Panel, increase the MINIMUMS knob and set Radar Altimeter (RA) Decision Height (DH) to 100 ft.

k) Verify a semi-transparent window is displayed in the upper corner of the LH/RH PFD.

l) On the Flight Control Panel, increase the MINIMUMS knob and set Radar Altimeter (RA) Decision Height (DH) to 150 ft.

m) Verify a MIN annunciation is now displayed in the semi-transparent window in the upper corner of the LH/RH PFD. Verify the word MINIMUMS is annunciated through the aural alerting system.

n) Decrease the Air Data Test set airspeed to 0 Knots. Ensure aircraft changes from “Air” to “Ground” mode.

o) Remove Air Data Test set from the aircraft pitot ports.

p) Turn aircraft power OFF. Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

q) Turn off and remove the DC variable power supply signal from the transceiver test input.

3) Height Calibration of RAD ALT (KR 405B)

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait several minutes for the aircraft electrical and avionics systems to initialize.

c) Ensure that RAD ALT 1 (J5) or RAD ALT 2 (J6) circuit breaker on the avionics circuit breaker panel is closed.

d) Wait two minutes to allow the RAD ALT to stabilize.

e) Verify that the radar altitude is shown on the LH/RH PFD. Make a note of the altitude displayed.

f) Locate the internal calibration switch hole on the connector side of the KR 405B transceiver.

g) Use a insert/extract tool (P/N M15570–20) to press the internal calibration switch. Hold the switch for 10 sec.

h) Verify on the LH/RH PFDs that the radar altitude displayed is 0 feet.

i) Turn aircraft power OFF. Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Radar Altimeter Antenna (DMPN 3–3A Pre Mod 6/2258, 071–01501–0000 Post Mod 6/2258)

1) General

These procedures are for the verification of the electrical bonding of the RAD ALT 1 antenna and RAD ALT 2 antenna, if installed.

2) Electrical Bonding Testing – Radar Altimeter Antenna

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Gain access to empennage through rear baggage compartment.

c) Locate Radar Altimeter Antenna forward of station 421.00 on the floor of the aircraft.

d) Verify that the electrical conductivity is consistent over the electrical bond contact area by placing a probe on the Rad Alt antenna mounting screw. Place the second probe on the nearest ground stud. The resistance between the two points must be < 3.0 milliohms unless stated otherwise.

e) Check the electrical conductivity between several random mounting screws within the electrical bond contact area to ensure consistency.

f) If the electrical conductivity is not consistent over the electrical bond contact area check the antenna electrical bonding in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

4. Removal/Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – ALL RACK MOUNT / REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MATING SURFACES BETWEEN THE LRU MOUNTING TRAY (OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO BONDING REQUIREMENTS IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE MOUNTING TRAY OR LRU CASE AS PER REQUIREMENTS IN PSM 1-64-2W.

A. Removal/Installation RAD ALT Transceiver (KR 405B)

1) General

These procedures are for the removal and installation of the RAD ALT 1 Transceiver and RAD ALT 2 Transceiver, if installed.

2) Removal

a) Ensure the aircraft power is OFF (refer to 24-00-00, Electrical Power – General Data).

b) Pull the RAD ALT 1 (J5) or RAD ALT 2 (J6) circuit breaker on the avionics circuit breaker panel and fit safety clip.

c) Gain access to the rear avionic shelves through the rear baggage compartment.

d) Disconnect the electrical and coaxial cables on the radar altimeter transceiver unit.

e) Loosen retaining screw clamp from the front lip of the transceiver and slide the unit out of the rack mount.

f) Remove unit from aircraft.

Note Stow connectors as required.

3) Installation

a) Ensure the aircraft power is OFF (refer to 24-00-00, Electrical Power – General Data).

b) Ensure the RAD ALT 1 (J5) or RAD ALT 2 (J6) circuit breaker on the avionics circuit breaker panel is pulled and fitted with safety clip.

c) Align the radar altimeter transceiver mounting holes to the bottom of the middle avionics shelf. Ensure correct orientation, connectors facing forward.

d) Slide the unit onto the rack mount. Tighten the retaining screw clamp at the front lip of the transceiver.

e) Reconnect the electrical and coaxial cable to the radar altimeter transceiver unit. Ensure connections are secure.

f) Remove safety clip and close RAD ALT 1 (J5) circuit breaker on the avionics circuit breaker panel.

g) Test the radar altimeter system as per Para 3.A.2).

B. Radar Altimeter Antenna (DMPN 3–3A Pre Mod 6/2258, 071–01501–0000 Post Mod 6/2258)

1) General

These procedures are for the removal and installation of the RAD ALT 1 Antenna and RAD ALT 2 Antenna, if installed.

2) Removal

a) Ensure the aircraft power is OFF (refer to 24-00-00, Electrical Power – General Data).

b) Pull the RAD ALT 1 (J5) or RAD ALT 2 (J6) circuit breaker on the avionics circuit breaker panel and fit safety clip.

c) Carefully cut the sealant from around the base of the antenna.

d) Remove the eight screws from the base of the antenna.

e) Carefully lift the antenna clear of the fuselage until there is access to the coaxial connection.

f) Disconnect the coaxial connector.

Note Stow connectors as required.

3) Installation

a) Ensure the aircraft power is OFF (refer to 24-00-00, Electrical Power – General Data).

b) Ensure the RAD ALT 1 (J5) or RAD ALT 2 (J6) circuit breaker on the avionics circuit breaker panel is pulled and fitted with safety clip.

c) Solvent clean the electrical contact bond areas immediately before the installation (refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

d) Bring the antenna assembly close to the fuselage, connect the coaxial connector.

e) Secure the antenna assembly on fuselage with the eight mounting screws.

f) Solvent clean the exposed area of airframe around the antenna (refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

g) Touch up the exposed area of airframe around the antenna if required with F19 primer by brush within 24 hours of the antenna installation. Ensure that the primer re-finishing provides complete coverage of the affected area.

h) If the affected area requires paint touch up, brush coat the area with the top-coat paint system matching the existing.

i) Use MIL–PRF–81733 sealant to both fillet seal around the antenna base and seal the securing screws. The aircraft will be ready to fly approximately 24 hours after sealing.

j) Check the antenna electrical bonding as per Para 3.B.2).

k) Remove safety clip and close RAD ALT 1 (J5) circuit breaker or RAD ALT 2 (J6) on the avionics circuit breaker panel.

l) Test the radar altimeter system as per Para 3.A.2).

C. Removal/Installation RAD ALT Mounting Tray

1) General

These procedures are for the removal and installation of the RAD ALT 1 Mounting Tray and RAD ALT 2 Mounting Tray, if installed.

2) Removal

a) Ensure the aircraft power is OFF (refer to 24-00-00, Electrical Power – General Data).

b) Gain access to the rear avionic shelves through the rear baggage compartment.

c) Loosen retaining screw clamp from the front lip of the transceiver and slide the unit out of the tray mount.

d) Remove the four screws that secure the mounting rack to the bottom of the rear avionics rack.

e) Remove mounting tray from aircraft.

3) Installation

a) Ensure the aircraft power is OFF (refer to 24-00-00, Electrical Power – General Data).

b) Gain access to the rear avionic shelves through the rear baggage compartment.

c) Align the mounting holes to the avionics shelf. Ensure correct orientation, retaining screw clamp facing forward.

d) Install the four screws and secure the mounting rack to the bottom of the rear avionics rack.

5. Troubleshooting

A. RAD ALT 1 and RAD ALT 2 Transceiver

1) If the RAD ALT is deemed inoperative check the following:

a) Ensure RAD ALT 1 (J5) or RAD ALT 2 (J6) circuit breaker is closed.

b) Visually inspect RAD ALT connector for damaged wires or contacts.

c) Visually inspect RAD ALT antenna and cable connection for corrosion and contaminants.

d) Ensure that the RAD ALT antenna coaxial and cable connector are properly mated and secure.

e) Inspect antenna external surfaces for damage.

f) Check that the antenna is properly mounted and screws are tight.

g) If troubleshooting at the wire level is required refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

h) If repair or service is required, send to appropriate rated avionics repair facility.